**COURSE: BCA (4TH SEMESTER)**
**Numerical Analysis and Scientific Computing -401**

*Block I: Overview & Roots of equation*

## 1: What is Floating Point Representation?

Floating point representation is a method of representing real numbers in scientific notation form:
Where:
m = mantissa
b = base
e = exponent
It is used to store very large or very small numbers in computers.

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## 2: What is Finite Precision?

Finite precision means that computers can store only limited digits of a number, causing small errors in calculations.

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## 3. What is Truncation Error?

Truncation error occurs when an infinite mathematical process (such as a series expansion or numerical method) is terminated after a finite number of steps. It is the difference between the exact mathematical result and the approximate result obtained by stopping early.

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# 4. What is Rounding Error?

Rounding error occurs because computers store numbers with limited precision. When a number is rounded to fit into memory, the small difference between the exact value and stored value is called rounding error.

## 5. Define Stability.

A numerical method is said to be stable if small errors (such as rounding errors) do not increase significantly during the computation process.

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## 6. Define Convergence.

A numerical method is convergent if the sequence of approximations approaches the exact solution as the number of iterations increases.

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## 7. Define Consistency.

A numerical method is consistent if the truncation error approaches zero as the step size approaches zero.

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## 8. Explain the Bisection Method.

The Bisection Method is an iterative method used to find the root of a continuous function in an interval [a,b]where $f(a)\,f(b)<0$.

Formula:

$$c = \frac{a+b}{2}$$

Steps:

1. Choose an interval [a,b] such that $f(a)f(b)<0$.
2. Find midpoint $c = \frac{a+b}{2}$
3. If $f(a)\,f(c)<0$, root lies in [a,c]; otherwise in [c,b].
4. Repeat the process until required accuracy is obtained.

The method is always convergent but slow.

## 9. Explain the False Position (Regula Falsi) Method.

The False Position Method is an iterative method that improves the Bisection Method by using linear interpolation.

Formula:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}$$

Steps:

1. Choose interval [a,b] such that $f(a)\,f(b)<0$.
2. Compute new approximation using the formula.
3. Replace either **a** or **b** based on sign.
4. Repeat until desired accuracy.

This method converges faster than Bisection in most cases.

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## 10. Explain the Newton–Raphson Method.

Newton–Raphson Method is a fast iterative method used to find the root of an equation using derivatives.

Formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f(x_n)}$$

Steps:

1. Choose an initial approximation $x_0$.
2. Substitute into the formula.
3. Repeat the process until successive values are close enough.

It converges rapidly but requires calculation of derivative.

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## 11. Compare Bisection and Newton–Raphson Methods.

| Bisection Method | Newton–Raphson Method |
|---|---|
| ➢ Slow convergence | ➢ Fast convergence |
| ➢ Does not require derivative | ➢ Requires derivative |
| ➢ Always convergent | ➢ May fail for poor initial guess |
| ➢ Simple to apply | ➢ Slightly complex |

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## 12. How are Polynomial Equations Solved Numerically?

A polynomial equation of the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = 0$$

can be solved using numerical methods such as:

- Newton–Raphson Method
- Iterative methods
- Synthetic division methods

These methods provide approximate roots.

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## 13. Explain Gaussian Elimination Method.

Gaussian Elimination is a method used to solve a system of linear equations.

Steps:

1. Write equations in matrix form.
2. Convert the coefficient matrix into an upper triangular matrix using row operations.
3. Use back substitution to find the values of unknowns.

It is widely used for solving large systems of equations.

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## 14. What is Pivoting?

Pivoting is the process of interchanging rows in Gaussian Elimination to avoid division by very small numbers and improve numerical stability.

Types:

- Partial Pivoting (row interchange)
- Complete Pivoting (row and column interchange)

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# 15. What is Floating Point Representation?

Floating point representation is a method used in computers to represent real numbers in the form:

$$x = \pm m \times b^e$$

where

- ***m*** = mantissa
- ***b*** = base
- ***e*** = exponent

It allows representation of very large and very small numbers using limited memory.

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# 16. What are Absolute Error and Relative Error?

**Absolute Error:**
The absolute difference between the true value and approximate value.

$$\text{Absolute Error} = |\text{True Value - Approximate Value}|$$

**Relative Error:**
The ratio of absolute error to the true value.

$$Relative\ Error = \frac{Absolute\ Error}{True\ Value}$$

Relative error gives the accuracy of approximation.

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# 17. What is Order of Convergence?

Order of convergence indicates how fast a numerical method approaches the exact solution.

If the error decreases rapidly in successive iterations, the method has higher order of convergence.

For example:

- Bisection method → Linear convergence
- Newton–Raphson method → Quadratic convergence

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# 18. Derive the Newton–Raphson Formula.

Start with Taylor series expansion of $f(x)$ about $x_n$

$$f(x) = f(x_n) + (x - x_n)f'(x_n)$$

If $x$ is root, then $f(x) = 0$.

So,

$$0 = f(x_n) + (x_{n+1} - x_n)f'(x_n)$$

Rearranging:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This is the Newton–Raphson formula.

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# 19. What are the Advantages and Disadvantages of Bisection Method?

**Advantages:**

- Simple to understand
- Always convergent
- Does not require derivative

**Disadvantages:**

- Slow convergence
- Requires interval where sign changes

## 20. What are the Advantages and Disadvantages of Newton–Raphson Method?

**Advantages:**

- Very fast convergence
- Requires fewer iterations

**Disadvantages:**

- Requires derivative
- May diverge if initial guess is poor
- Not suitable when derivative is zero

*Block II: Curve Fitting, Differentiation & Integration*

## 1. What is Curve Fitting?

Curve fitting is a method of finding a mathematical function that best fits a given set of data points.

It is used to approximate the relationship between variables.

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## 2. Explain the Method of Least Squares.

The method of least squares is used to find the best fitting curve by minimizing the sum of squares of errors between observed and calculated values.

For straight line:

$y = a + bx$

Normal equations:

$$\sum y = na + b\sum$$

$$\sum xy = a\sum x + b\sum$$

Solve these equations to find ***a*** and ***b***.

## 3. What is Interpolation?

Interpolation is a method used to estimate the value of a function for a given value of ***x*** within the range of known data points.

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## 4. What are Finite Differences?

Finite differences are used to approximate derivatives and construct interpolation formulas.

Types:

- Forward difference
- Backward difference
- Central difference

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## 5. State Newton's Forward Interpolation Formula.

Used when data points are equally spaced and value lies near beginning.

$$f(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \cdots$$

Where

$$u = \frac{x - x_0}{h}$$

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## 6. State Newton's Backward Interpolation Formula.

Used when value lies near the end of table.

$f(x) = y_n + u\nabla y_n + \frac{u(u+1)}{2!}\nabla^2 y_n + \cdots$

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## 7. What is Lagrange's Polynomial Interpolation Formula?

Used for unequal intervals.

$$f(x) = \sum_{i=0}^{n} y_i L_i(x)$$

Where $L_i(x)$ are Lagrange basis polynomials.

It does not require difference table.

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## 8. What is Newton's Divided Difference Formula?

Used for unequal intervals.

General form:

$$f(x) = f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \cdots$$

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## 9. What is Numerical Differentiation?

Numerical differentiation is the process of approximating the derivative of a function using discrete data points.

Example:

$f'(x) \approx \frac{f(x+h) - f(x)}{h}$

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## 10. What is Numerical Integration?

Numerical integration is the process of approximating the definite integral of a function when exact integration is difficult.

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## 11. State the Trapezoidal Rule.

$$\int_a^b f(x)dx \approx \frac{h}{2}[y_0 + y_n + 2(y_1 + y_2 + \cdots + y_{n-1})]$$

It approximates area using trapeziums.

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## 12. State Simpson's 1/3 Rule.

$$\int_a^b f(x)\,dx \approx \frac{h}{3}[y_0 + y_n + 4(y_1 + y_3 + \cdots) + 2(y_2 + y_4 + \cdots)]$$

More accurate than trapezoidal rule.

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## 13. What is Taylor's Series Method?

Taylor's method is used to solve differential equations by expanding function into Taylor series.

$$y(x+h) = y(x) + hy'(x) + \frac{h^2}{2}y''(x) + \cdots$$

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## 14. What is Euler's Method?

Euler's method is used to solve first-order differential equations.

Formula:

$$y_{n+1} = y_n + hf(x_n, y_n)$$

It is simple but less accurate.

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## 15. What is Modified Euler's Method?

Improved version of Euler's method.

$$y_{n+1} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_{n+1,} y_{n+1})]$$

More accurate than simple Euler.

## 16. What is Picard's Method?

Picard's method is an iterative method used to solve differential equations using successive approximations.

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## 17. What is Runge–Kutta Method?

Runge–Kutta methods are higher order methods used to solve differential equations.

Fourth order Runge–Kutta (RK4) is most commonly used and gives very accurate results.

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## 18. Compare Trapezoidal Rule and Simpson's Rule.

| Trapezoidal | Simpson's |
|---|---|
| ➢ Uses straight lines | ➢ Uses parabolic arcs |
| ➢ Less accurate | ➢ More accurate |
| ➢ Simpler | ➢ Slightly complex |

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## 19. What are the Advantages of Least Squares Method?

- Gives best fit curve
- Minimizes error
- Widely used in data analysis

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## 20. Differentiate Between Interpolation and Extrapolation.

| Interpolation | Extrapolation |
|---|---|
| ➢ Estimate within data range | ➢ Estimate outside data range |
| ➢ More reliable | ➢ Less reliable |